



Predictive Modeling of Airline Route Entry Decisions with a Logit Based Framework

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Agenda

1. Introduction
2. Methods
3. Results
4. Next Steps



Introduction

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Air transport is a key driver of economic activity and social mobility in the Philippines.

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- Inter-island air connectivity supports tourism, business links, and labor mobility.
- **Cebu Pacific (Cebu Air, Inc.)** is the largest domestic operator by flight count and network breadth, annually carrying millions of passengers while executing major fleet expansions. [5][6][7][8][20]

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- Public reporting and airline disclosures document rapid increases in annual seat capacity and expanded market share.
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- Public reporting and airline disclosures document rapid increases in annual seat capacity and expanded market share.
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- Expanded capacity enables testing of new service patterns across more city pairs but raises managerial risk when allocating finite aircraft and commercial resources.

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- **Objective and data-driven route-selection methods** help airlines and external stakeholders make defensible deployment choices.

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 - Construct a panel dataset of market, network, and competition features and estimate a binary logit model.
 - Evaluate the model and apply the fitted model to prioritize unserved routes.

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1. What **route-level and airport-level factors** most influence the likelihood that a low-cost carrier will successfully establish and sustain a new domestic route in the Philippines?
2. To what extent **can a logit model predict the binary outcome of route success** (viable/not viable) using historical route and market data?

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4. How sensitive are the model predictions to **changes in key variables**?
5. What **practical recommendations** for route prioritization can be derived from the model results?

Assumptions and Limitations of the Study

- The focus of the study is on **domestic Philippine routes** only.
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- **Manual extraction** from Flightradar24 constrains temporal coverage and may introduce omissions. [17]
- **Unobserved factors** such as contractual airport agreements, slot negotiations, exact aircraft availability windows, and corporate financial objectives are not modeled.

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- Assumes that past behavior is **consistent**; structural changes after the data period may reduce predictive validity.

Discrete-choice models effectively explain airline route-entry decisions.

- **Discrete-choice (logit/probit) frameworks** treat each candidate city-pair as a binary choice and are widely validated in route-entry research. [26][19][32]

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- **Discrete-choice (logit/probit) frameworks** treat each candidate city-pair as a binary choice and are widely validated in route-entry research. [26][19][32]
- **Key predictors** consistently include market size, demand proxies, distance/friction, endpoint connectivity, and competition metrics. [26][32][19][23]

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- Best practice is to **combine regulatory and tourism statistics with commercial flight data** to build panel datasets suitable for discrete-choice estimation. [26][32]
- **Methodological gaps of past studies:** few studies target Southeast Asian/archipelagic domestic markets, many use annual (not monthly) panels, and few integrate both regulator data and detailed operational traces. [23][26][32]



Methods

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Data Collection

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- Duplicate records and observations missing essential identifiers were removed to guarantee dataset integrity.

Summary of Features

Route Variables	Airport Variables	Socioeconomic Variables (To & From)
- Distance - Estimated passengers - Route-level HHI	- Airport size (To & From) - Airline own share of flights - Airport-level HHI (To & From)	- Count of business and professional organizations. - Accommodation capacity (rooms/beds). - Distance to nearest port (km). - Local economic growth indicator. - Number of accommodation and food-service establishments. - Count of households with OFWs. - Count of overnight tourist arrivals reported by accommodations.

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- **Estimated passenger counts** per flight were computed as aircraft certified seat capacity multiplied by airline-period load factor and then aggregated to the route level.

Route Variables

Estimated Passengers: Example

- Given A320 with 180 seats and an average load factor of 0.78, then estimated passengers = $180 \times 0.78 = 140.4 \approx 140$
- If ten such arrivals are observed for route r in the sample period, then we have $140 \times 10 = 1400$.

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- Larger HHI values indicate greater concentration (fewer firms accounting for a larger portion of the market), while smaller values indicate a more competitive market structure.

Route Variables

Route-level HHI (RouteHHI)

Market concentration on route r is measured by the Herfindahl-Hirschman Index using seat shares:

$$\text{RouteHHI}_r = \sum_a (s_{a,r})^2, \quad s_{a,r} = \frac{\text{seats}_{a,r}}{\sum_b \text{seats}_{b,r}}$$

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- Route-level HHI quantifies market concentration specific to the origin-destination pair.
- Larger values indicate markets dominated by few carriers while smaller values indicate more fragmented competition.

Route Variables

Route-level HHI (RouteHHI): Example

- Suppose that Philippine Airlines has 10,000 seats (67% market share) and Cebu Pacific has 5,000 seats (33% market share) for a specific route.
- $\text{RouteHHI} = 67^2 + 33^2 = 4,489 + 1,089 = 5,578$

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- **Airline own share on a route** is measured as the airline's proportion of seats (or estimated passengers) on route r .

Airport Variables

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- Consider a route where Cebu Pacific operates two A320 flights, each with 180 seats, while other carriers operate a single ATR72 with 70 seats.

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- Consider a route where Cebu Pacific operates two A320 flights, each with 180 seats, while other carriers operate a single ATR72 with 70 seats.
- Cebu Pacific therefore supplies 360 seats on that route (2×180), while the other carriers supply 70 seats, making the route's total seating capacity 430 ($360 + 70$) seats.

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- Cebu Pacific therefore supplies 360 seats on that route (2×180), while the other carriers supply 70 seats, making the route's total seating capacity 430 ($360 + 70$) seats.
- Cebu Pacific's share of seats is 360 out of 430, which is approximately 83.7%.

Airport Variables

Airport-level HHI (AirHHI)

For airport i , an HHI that pools all flights touching i was computed. Let $S_{a,i}$ be airline a 's share of seats (or passengers) at airport

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- AirHHI is distinct from RouteHHI because it reflects the broader competitive environment faced by an airline at a hub or gateway.

Airport Variables

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- Philippine Airlines has 80,000 seats across all MNL routes (60%), Cebu Pacific has 50,000 seats across all MNL routes (37.5%), and other airlines have 3,333 seats (2.5%).
- $\text{AirHHI} = 60^2 + 37.5^2 + 2.5^2 = 3,600 + 1,406 + 6.25 = 5,012.25$

Socioeconomic Variables

Cities and Municipalities Competitive Index (CMCI)

- Annual ranking of Philippine cities and municipalities on different "pillars":
 - Economic Dynamism
 - Government Efficiency
 - Infrastructure
 - Resiliency
 - Innovation
- Provides us with a metric for the quality of life, governance, and business environment of a place.

Socioeconomic Variables (CMCI)

From CMCI, the following variables were extracted:

- **Presence of business and professional organizations** - proxy for local business activity
- **Accommodation capacity** - proxy for tourists and VFR travelers
- **Distance to the nearest port** - proxy for multimodal accessibility
- **Measures of local economic growth** - proxy for expanding income and business activities

Socioeconomic Variables (PSA)

From the Philippine Statistics Authority, the following variables were used:

- **Counts of accommodation and food-service establishments** - proxy for tourists and VFR travelers
- **Counts of households with overseas Filipino workers (OFWs)** - proxy for VFR-related mobility

Socioeconomic Variables (DOT)

From the Department of Tourism, the following variable was utilized:

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Endpoint airports' socioeconomic attributes (both origin and destination) were likewise incorporated into the modelling dataset.

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Logistic (Logit) Model

$$p_i = \frac{1}{1 + \exp(-\beta^\top X_i)} = \text{logit}^{-1}(\beta^\top X_i)$$

where $X_i \in \mathbb{R}^p$ is a vector of covariates and $\beta \in \mathbb{R}^p$ is a parameter vector.

Logit Regression: Route Entry Prediction

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Route Entry Prediction

$$\text{Viable}_i = \begin{cases} 1 & \text{if } p_i(\hat{\beta}) \geq 0.5, \\ 0 & \text{if } p_i(\hat{\beta}) < 0.5. \end{cases}$$

where $p_i(\hat{\beta}) \in [0, 1]$ is the probability for each observation.

Model Evaluation

- The logit model's performance was assessed through the evaluation metrics:
 - Brier Score
 - The area under the receiver operating characteristic curve (ROC AUC)
 - McFadden's pseudo-R-squared

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Brier Score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - Y_i)^2$$

where n is the number of all observations, p_i is the fitted probability from the logit model, and $Y_i \in \{0, 1\}$ is the observed entry indicator.

Model Evaluation: ROC AUC

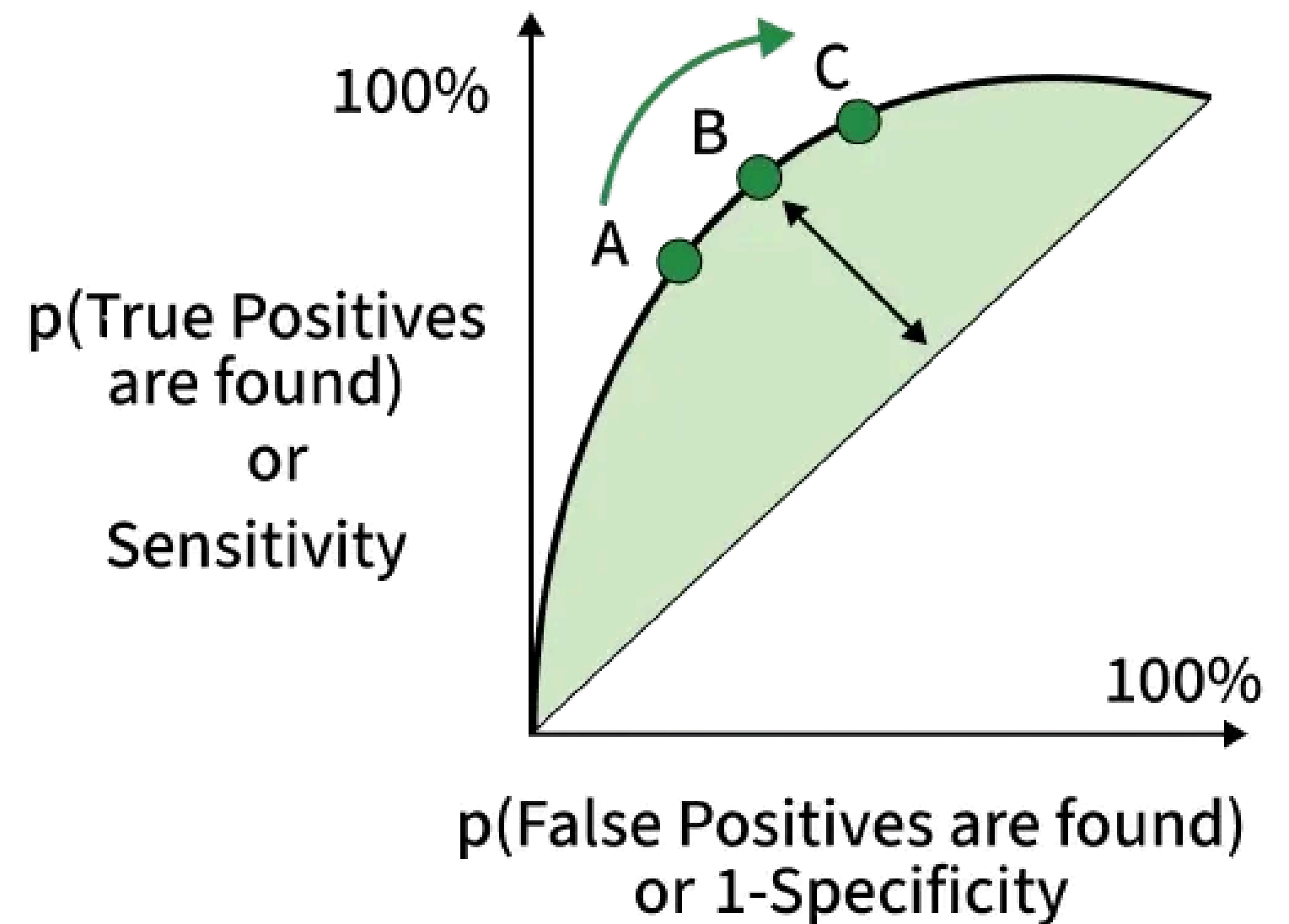
- ROC AUC measures how well the model ranks candidate routes (probability that a true-entry route is scored higher than a non-entry route).

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- Values above 0.7 generally denote acceptable discrimination.

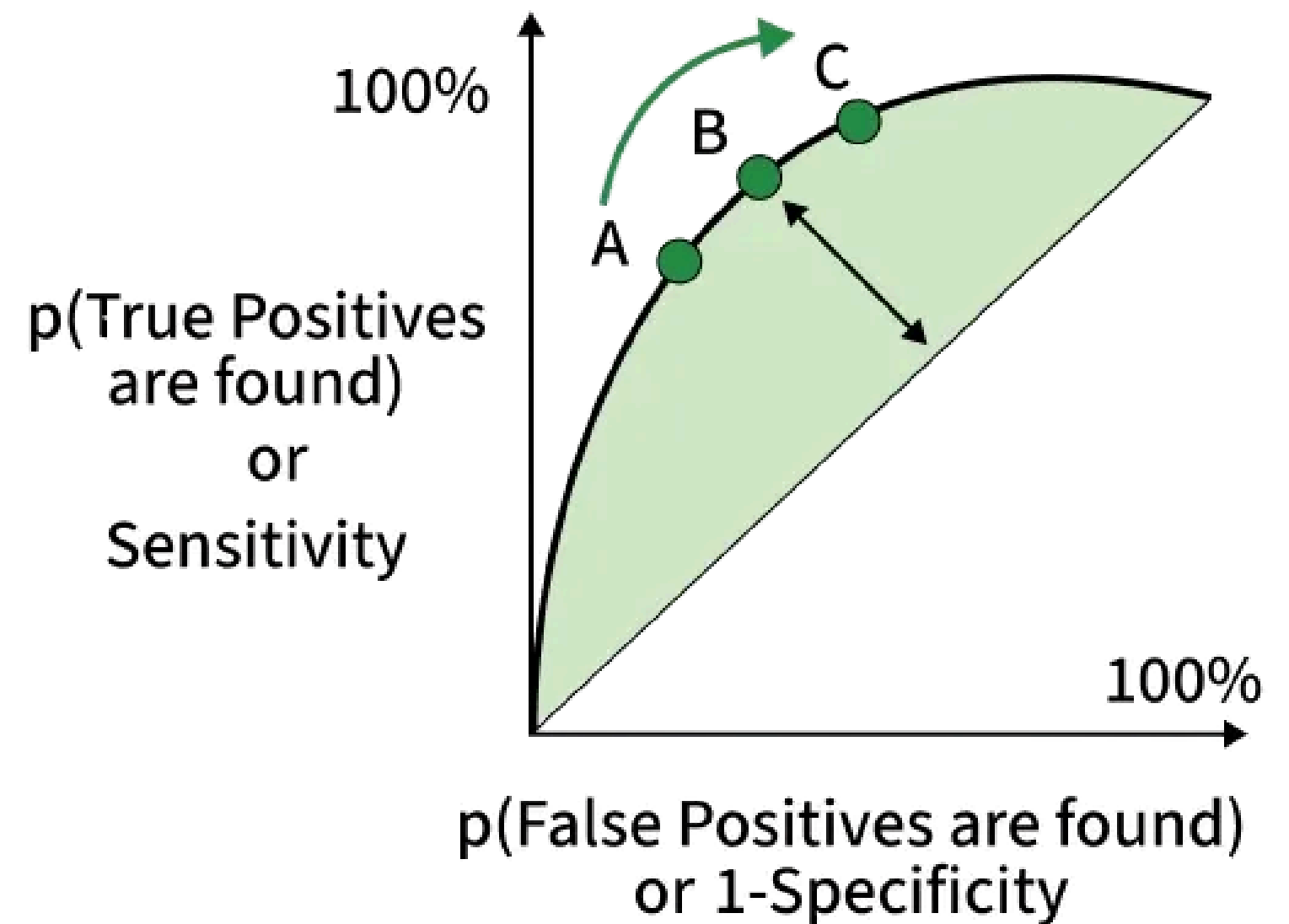
Model Evaluation: ROC AUC

- False Positive Rate (FPR): how often the model incorrectly predicts a negative case as positive.



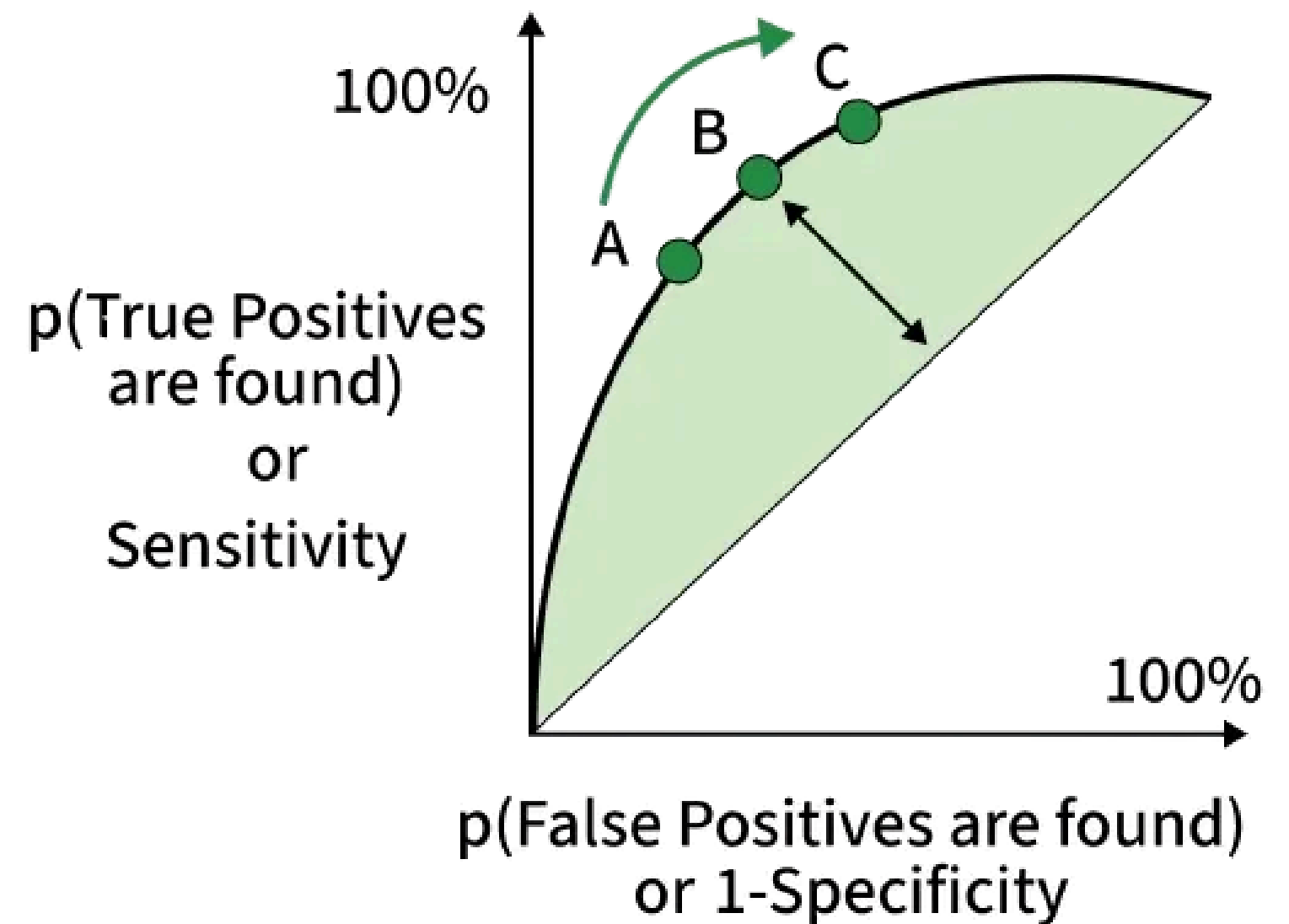
Model Evaluation: ROC AUC

- False Positive Rate (FPR): how often the model incorrectly predicts a negative case as positive.
- True Positive Rate (TPR): how often the model correctly predicts the positive cases.



Model Evaluation: ROC AUC

- The diagonal line from the bottom-left to the top-right is the baseline: a model that ranks at random would fall along this line.



Model Evaluation: Pseudo-R-squared

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McFadden's Pseudo-R-squared

Let ℓ_{model} be the log-likelihood of the fitted model and ℓ_{null} the log-likelihood of the intercept-only model. Then

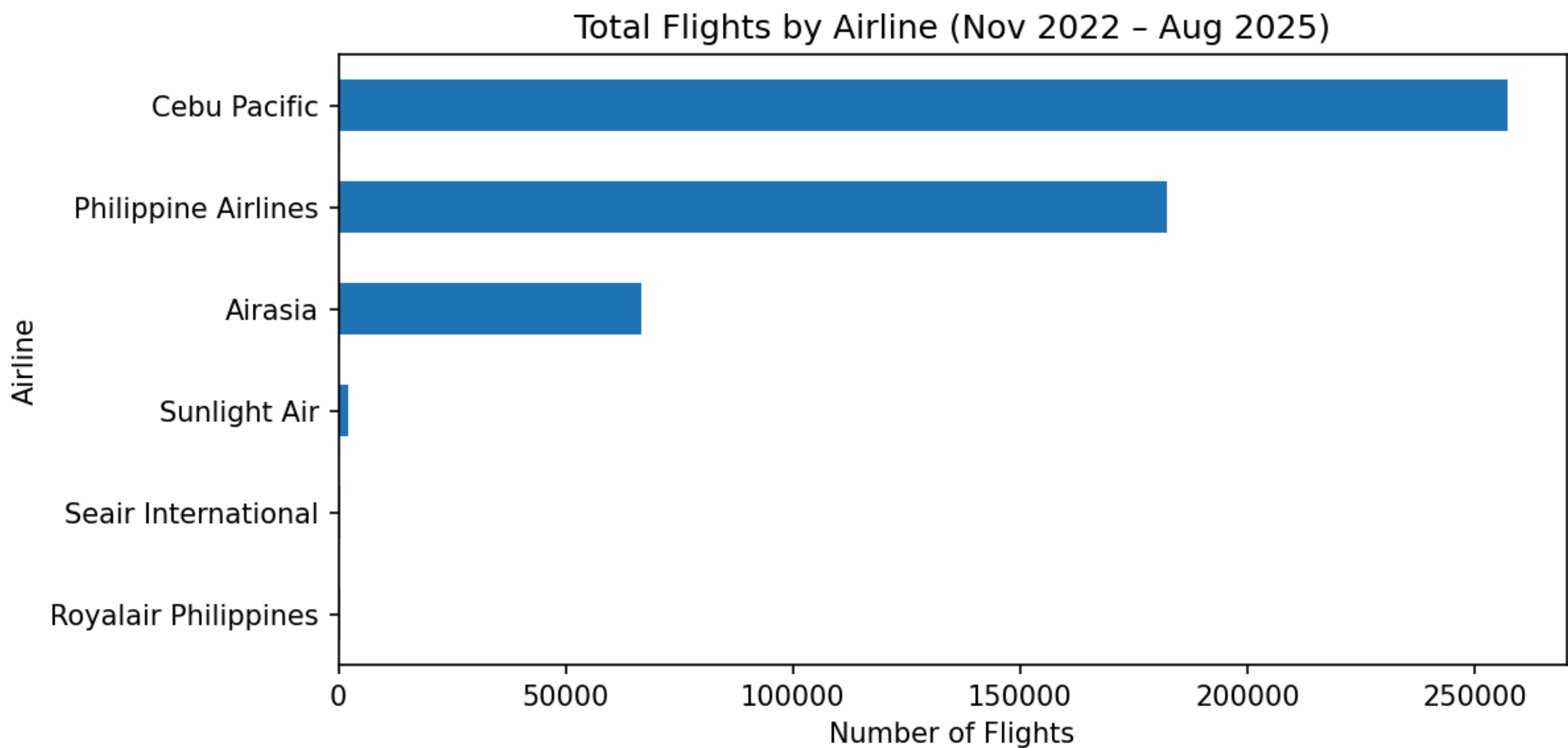
$$R^2_{\text{McF}} = 1 - \frac{\ell_{\text{model}}}{\ell_{\text{null}}}$$



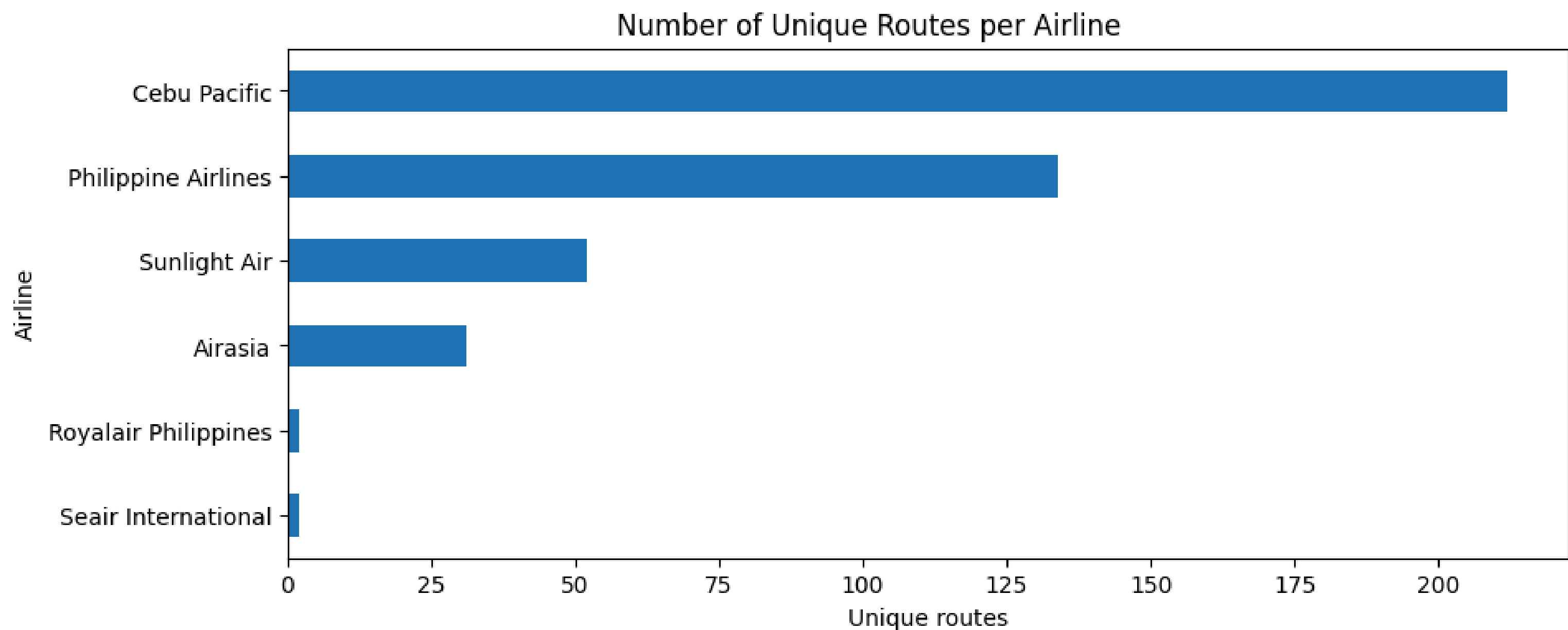
Results

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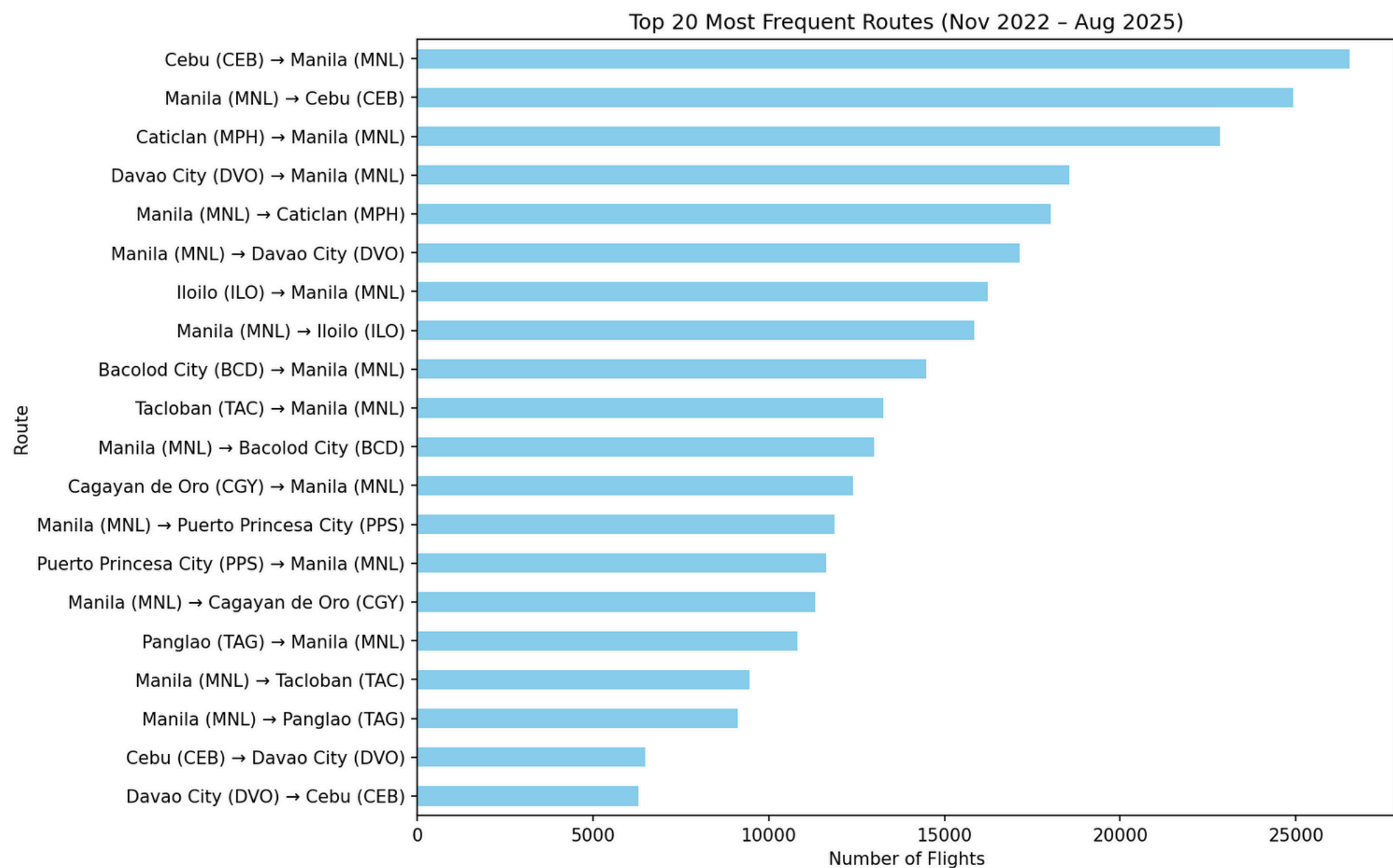
Cebu Pacific group operated the largest share of domestic flights.



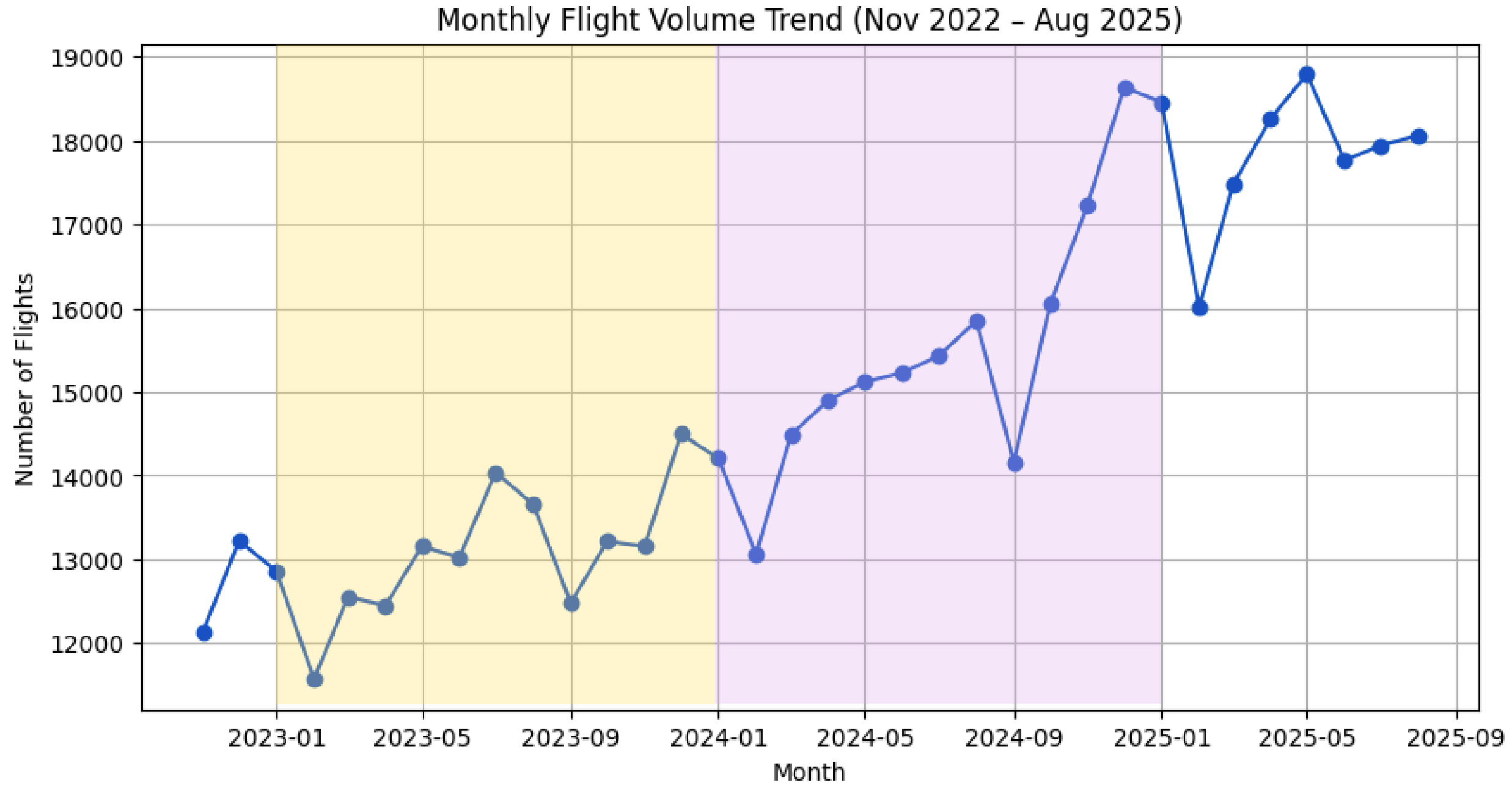
Cebu Pacific maintains the widest domestic network.



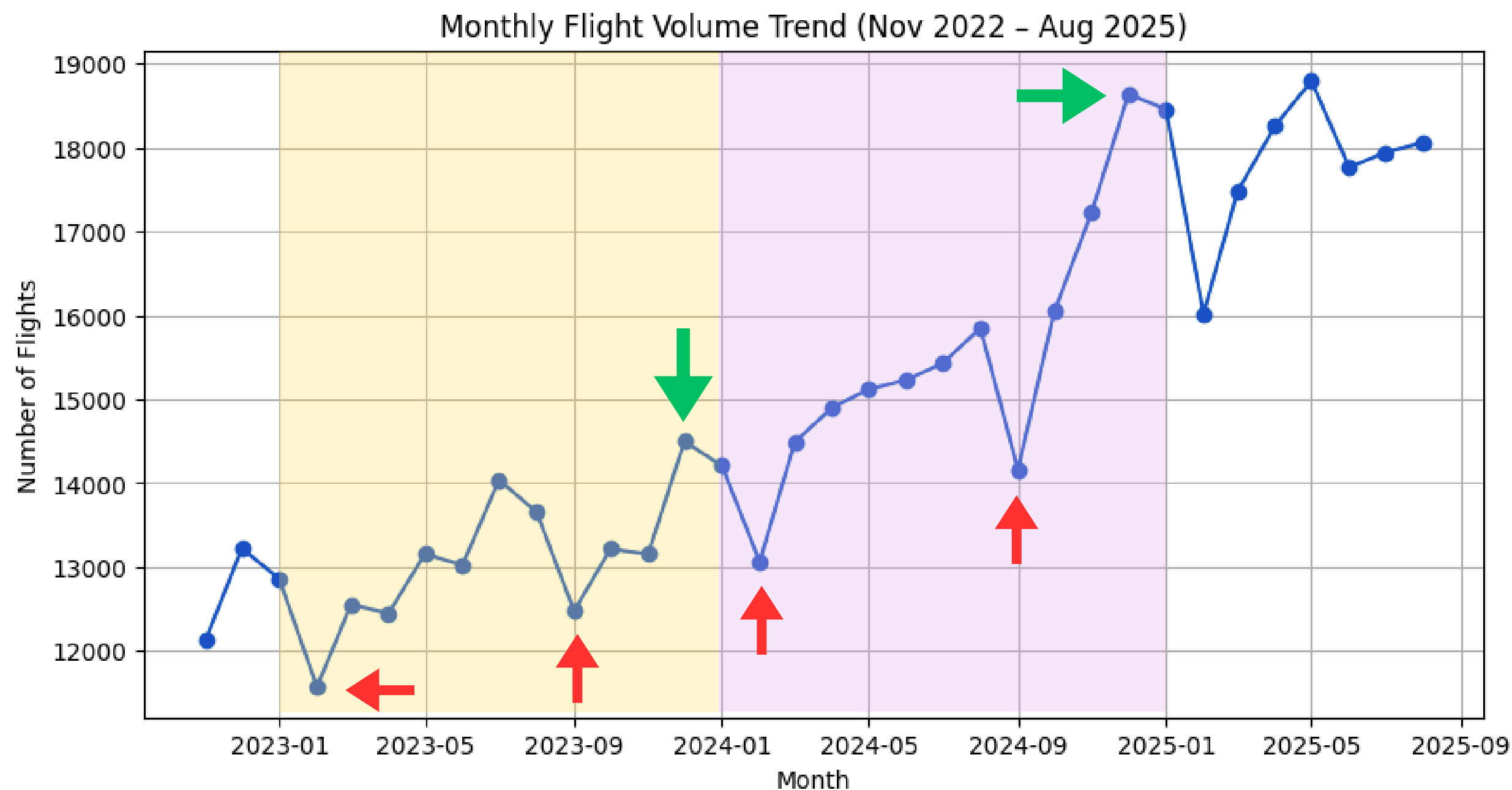
18 of the top 20 routes involve MNL.



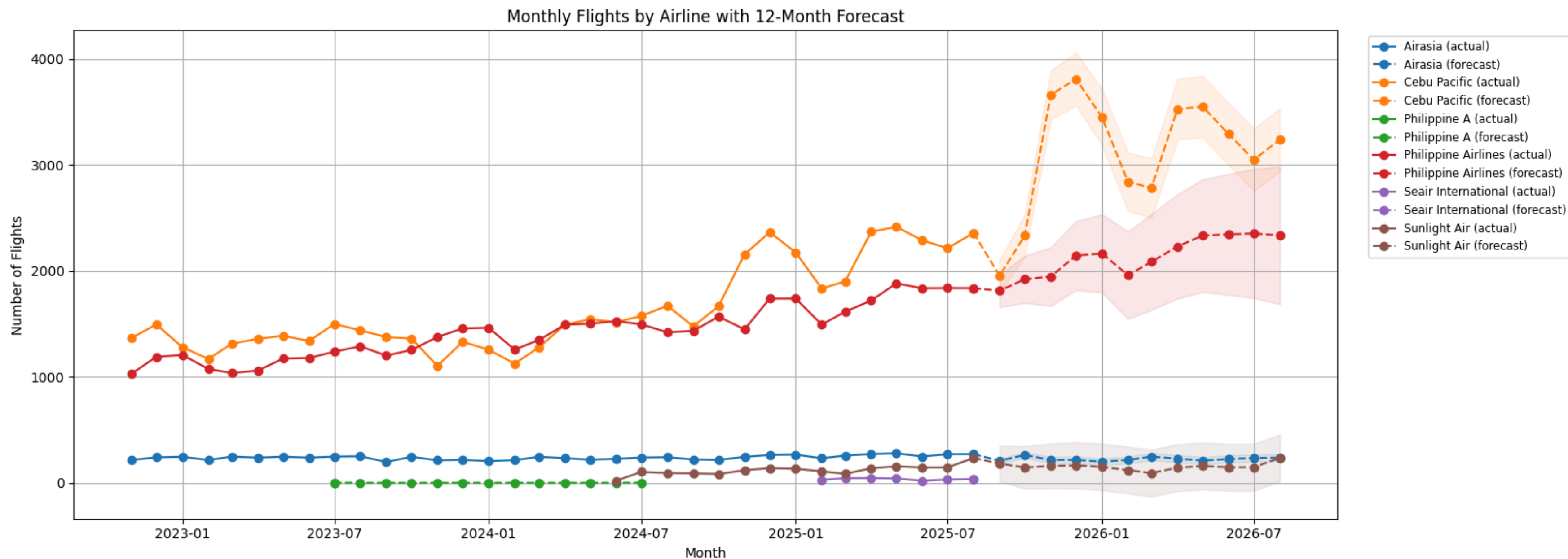
A seasonal pattern is observed for both 2023 and 2024.



Peaks in December and summer; Falls in February and September.



Demand is expected to increase in 2026.



Logit modeling over 2024 data is more robust.

	Metric	Full Period	2024 Only
	Observations	7955	2580
	Log-Likelihood	-2897.9	-844.30
Higher	Pseudo R ²	0.4735	0.5274
Lower	Brier Score	0.1076	0.0980
Higher	ROC AUC	0.9109	0.9290

Results show which variables have a significant impact on the model.

Variable	Coefficient	Std. Error	p-value
const	-0.2198	0.073	0.003
RouteHHI	2.7336	0.135	0.000
AirHHI_From	-1.0953	0.104	0.000
AirHHI_To	-1.1805	0.104	0.000
AirFli_From	1.8066	0.352	0.000
AirFli_To	2.7267	0.407	0.000
Business_To	-0.0660	0.080	0.411
Business_From	0.0691	0.081	0.392
Accommodation_From	-0.0063	0.088	0.943
Accommodation_To	-0.1941	0.087	0.026

Variable	Coefficient	Std. Error	p-value
Port_From	-0.4291	0.155	0.006
Port_To	-0.1035	0.117	0.377
EGrowth_From	1.1125	0.159	0.000
EGrowth_To	0.7427	0.135	0.000
OFW_From	-0.5741	0.291	0.049
OFW_To	0.3568	0.264	0.176
Travelers_From	1.1988	0.156	0.000
Travelers_To	1.1305	0.152	0.000
Establishment_From	-1.4789	0.428	0.001
Establishment_To	-2.8905	0.465	0.000

Insights From The Model (Positive Coefficients)

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- **Overnight Travelers (Travelers_From, Travelers_To)**
 - Higher tourism volume increases the likelihood of entry

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 - Larger accommodation capacity is associated with lower probability of entry
- **Establishment_From / Establishment_To**
 - Greater numbers of establishments are linked to lower entry likelihood

New route entry reflects Cebu Pacific's hub-and-spoke network structure.

	Route	Pred_Prob_2024	Pred_Entry_2024		Route	Pred_Prob_2024	Pred_Entry_2024
0	CEB-BAG	0.624284	1	16	CBO-TAG	0.000164	0
1	CYP-MNL	0.971236	1	17	CBO-TWT	0.001881	0
2	CYP-TAC	0.001113	0	18	CEB-BPA	0.697681	1
3	IAO-MNL	0.992039	1	19	CEB-CBO	0.808824	1
4	KLO-MPH	0.004666	0	20	CEB-CRM	0.680517	1
5	MNL-CYP	0.960066	1	21	CRK-BSO	0.289110	0
6	MPH-KLO	0.010058	0	22	CRM-CEB	0.740917	1
7	SUG-BXU	0.000418	0	23	CRM-MNL	0.750369	1
8	SUG-TAG	0.000147	0	24	DVO-CBO	0.409249	0
9	TBH-MNL	0.522772	1	25	EUQ-MNL	0.935602	1
10	TBH-MPH	0.028614	0	26	MNL-BSO	0.686557	1
11	BPA-CEB	0.777171	1	27	MNL-CBO	0.831233	1
12	BSO-CRK	0.250756	0	28	MNL-CRM	0.712621	1
13	BSO-MNL	0.004055	0	29	MNL-EUQ	0.949067	1
14	CBO-CEB	0.882429	1	30	TWT-CBO	0.001613	0
15	CBO-MNL	0.887502	1				

New route entry reflects Cebu Pacific's hub-and-spoke network structure.

- Manila-Cotabato
- Manila-Calbayog
- Manila-Siargao
- Manila-Alcantara
- Manila-Catarman
- Manila-Antique
- Manila-Basco
- Cebu-Borongan
- Cebu-Cotabato
- Cebu-Baguio
- Cebu-Catarman

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 - Allows testing whether flight supply adjustments influence Cebu Pacific's likelihood of entering or maintaining a route.

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- **Add schedule frequency or schedule changes as predictors**
 - Allows testing whether flight supply adjustments influence Cebu Pacific's likelihood of entering or maintaining a route.
- **Extend the model to include currently unserved routes**
 - Uses socioeconomic variables to identify high-potential but overlooked markets for future network development and policy support.



Thank You

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Predictive Modeling of Airline Route Entry Decisions with a Logit Based Framework

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Dizo, Lee, Yumol

Route Variables

Estimated Passengers

Let $\hat{P}_f$ denote the estimated number of passengers on flight f operating on route r

$$\hat{P}_f = C_a \times LF_{a,t},$$

where C_a is the certified maximum seat capacity for aircraft type a and $LF_{a,t}$ is the empirical load factor for airline a in period t .

Route Variables

Estimated Passengers (Continued)

Route-level demand is obtained by aggregating flight-level estimates across all recorded arrivals on route r within the sample window:

$$\hat{P}_r = \sum_{f \in r} \hat{P}_f$$

Airport Variables

- Airport size is defined as the annualized count of recorded arrivals at an airport.

Airport Size (AirFli)

$$\text{AirFli}_i = \sum_{r \in \mathcal{R}(i)} n_{r,i}$$

where $n_{r,i}$ is the number of recorded flights at airport i that are part of route r , and $\mathcal{R}(i)$ is the set of routes involving i .

Airport Variables

- Airline own share on a route is measured as the airline's proportion of seats (or estimated passengers) on route r .

Airline Own Share of Flights on Route (OwnShfli)

For airline a on route r , the airline's share of seats (or flights) can be calculated as

$$\text{OwnShfli}_{a,r} = \frac{\text{seats}_{a,r}}{\sum_b \text{seats}_{b,r}} \times 100$$

where $\text{seats}_{a,r}$ is the sum estimated passengers $\hat{P}_f$.

Model Evaluation: ROC AUC

Area Under the Receiver Operating Characteristic Curve (ROC AUC)

ID	Predicted probability	True label
A	0.90	1 (positive)
B	0.80	0 (negative)
C	0.60	1 (positive)
D	0.30	0 (negative)

Model Evaluation: ROC AUC

Area Under the Receiver Operating Characteristic Curve (ROC AUC)

- Identify positives and negatives.
 - Positives: A, C
 - Negatives: B, D

Model Evaluation: ROC AUC

Area Under the Receiver Operating Characteristic Curve (ROC AUC)

- Identify positives and negatives.
 - Positives: A, C
 - Negatives: B, D
- Form all positive-negative pairs and compare scores. There are $2 \times 2 = 4$ pairs:
 - Pair (A, B): 0.90 vs 0.80 → positive has higher score → score = 1
 - Pair (A, D): 0.90 vs 0.30 → positive higher → score = 1
 - Pair (C, B): 0.60 vs 0.80 → positive lower → score = 0
 - Pair (C, D): 0.60 vs 0.30 → positive higher → score = 1

Model Evaluation: ROC AUC

Area Under the Receiver Operating Characteristic Curve (ROC AUC)

- Sum the pair scores and divide by the number of pairs to get AUC:
 - $AUC = (1+1+0+1)/4 = 3/4 = 0.75$.

Model Evaluation: ROC AUC

Area Under the Receiver Operating Characteristic Curve (ROC AUC)

- Sum the pair scores and divide by the number of pairs to get AUC:
 - $AUC = (1+1+0+1)/4 = 3/4 = 0.75$.
- Interpretation: An AUC of 0.75 means that, if one positive route and one negative route were picked at random from this sample, there is a 75% chance the model assigns a higher predicted probability to the positive route than to the negative one.

Slight variation in seasonality between tourism- and business-driven routes.

